

1. A 13-year-old girl is brought to the physician for a routine health maintenance examination. She feels well but says she worries about her weight; she reports that for as long as she can remember, she has been taller and larger than the other kids in her class. She is a cheerleader at her middle school and describes herself as the heaviest girl on the squad. Her attempts to lose weight through diet have been unsuccessful. She usually eats three meals daily and tries to eat a lot of vegetables and fruit. She also exercises daily, often running or bicycling on days she does not have cheerleading practice. She reports that she is a good student and has many friends. Menarche was at the age of 11 years. Menses occur at regular 28-day intervals. She appears healthy and somewhat older than her stated age. She is 157 cm (5 ft 2 in) tall and weighs 52 kg (115 lb); BMI is 21 kg/m². Vital signs are within normal limits. Physical examination shows no abnormalities. Breast and genital development are sexual maturity rating stage 4. She has a full range of affect and normal thought process. Which of the following is the most likely diagnosis?
- ☐ A) Adjustment disorder with anxiety
 - ☐ B) Anorexia nervosa
 - ☐ C) Body dysmorphic disorder
 - ☐ D) Generalized anxiety disorder
 - ☐ E) Normal adolescence



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Pause

2. A 27-year-old man is brought to the office by his parents because of increasingly erratic behavior. He completed college without difficulty but moved back to his parents' home 5 years ago because he was having financial difficulties. He is an aspiring writer and has begun writing several novels and travel guides. One year ago, he began to collect junk mail and papers in his bedroom and would not allow his parents to discard any trash. He began to wear the same clothes for several days in a row. During the past month, he has refused to leave his bedroom and now spends all his time working on his computer. He says that he is doing internet research for his writing projects and sending e-mails to his friends. Last week, his parents discovered that he had not been connected to the internet for 3 months, and his e-mail account showed no activity for the past 4 months. He has no history of serious illness and takes no medications. He does not drink alcoholic beverages or use illicit drugs. He is poorly groomed and slightly overweight. Physical examination, including neurologic examination, shows no abnormalities. On mental status examination, most of his replies are "yes" or "no." He describes his mood as "okay" and has a flat affect. He says he is worried only about his computer work and requests that his parents not delete anything because his writing is "important" and "not ready." There is no evidence of hallucinations. Which of the following is the most likely diagnosis?
- ☐ A) Delusional disorder
 - ☐ B) Intellectual developmental disorder
 - ☐ C) Obsessive-compulsive disorder
 - ☐ D) Paranoid personality disorder
 - ☐ E) Schizophrenia

3. A 32-year-old woman comes to the office because of a 1-day history of headache, anxiety, nausea, muscle aches, and diarrhea. Ibuprofen has not fully resolved the headache. She was diagnosed with major depressive disorder 8 months ago, but she has discontinued the dosage of her sertraline during the past month because she is trying to conceive. She has not had any alcoholic beverages during the past 2 weeks; she previously consumed six glasses of wine weekly. She does not use any illicit drugs. Vital signs are within normal limits. Physical examination shows a mild tremor in both hands. On mental status examination, she describes her mood as "worried" and has a tired affect. Which of the following is the most likely explanation for this patient's symptoms?
- ☐ A) Alcohol use disorder
 - ☐ B) Influenza
 - ☐ C) Migraine headache
 - ☐ D) Pregnancy
 - ☐ E) Selective serotonin reuptake inhibitor withdrawal
 - ☐ F) Unspecified anxiety disorder



4. An 18-year-old man comes to the physician because of constant anxiety for 2 weeks. During this time, he has been unable to stop pacing around his parents' house. He describes his sleep as "okay." He has not had a decreased appetite but does find it difficult to sit still long enough to eat a meal. He is able to go to his summer job but has difficulty concentrating. He says, "I feel like I am crawling out of my skin." His girlfriend left for Europe 3 weeks ago, and he fears that they will grow apart and she will not return. His temperature is 36.8°C (98.2°F), pulse is 98/min, and blood pressure is 150/90 mm Hg. Physical examination shows no abnormalities. On mental status examination, he is agitated and unable to sit still. Which of the following is the most likely diagnosis?
- ☐ A) Acute stress disorder
 - ☐ B) Adjustment disorder with anxiety
 - ☐ C) Bipolar disorder
 - ☐ D) Dependent personality disorder
 - ☐ E) Generalized anxiety disorder
 - ☐ F) Major depressive disorder

5. A 52-year-old woman comes to the physician because of insomnia, dysphoria, and irritability for 3 weeks; during this period, she has had a high energy level. She has not been able to focus on her work and has taken numerous sick days. Although she had similar episodes in the past, the episodes have increased in severity and frequency over the past 2 years. She now has up to six episodes yearly. She currently takes no medications. She does not use illicit drugs. Examination shows no abnormalities. Which of the following is the most appropriate pharmacotherapy for this patient?
- ☐ A) Alprazolam
 - ☐ B) Amitriptyline
 - ☐ C) Buspirone
 - ☐ D) Divalproex
 - ☐ E) Fluoxetine



6. A 57-year-old woman comes to the office to obtain results of evaluation of lower abdominal pain. CT scan of the abdomen and pelvis shows a large ovarian mass highly suspicious for ovarian carcinoma as well as numerous small masses throughout the peritoneum. When informed of the findings, the patient expresses disbelief. A few minutes later, she begins to cry. She says that she does not know what she is going to do. Which of the following is the most appropriate response by the physician?
- ☐ A) "Let's plan on meeting again soon to discuss the situation."
 - ☐ B) "May I step out to get my nurse?"
 - ☐ C) "Please don't be upset. Everything will be ok."
 - ☐ D) "Try to compose yourself or we won't be able to provide you with proper care."
 - ☐ E) "We need to start discussing treatment options now."

7. A 62-year-old woman comes to the physician with her husband because of difficulty concentrating during the past 3 months. She says, "I can't think any more, and I can't stop shaking." She also has had difficulty sleeping, decreased level of energy, and a decreased appetite, resulting in a 6.3-kg (14-lb) weight loss. She has stayed home from work for the past 4 days because of her symptoms and worries that she may never be able to work again. She no longer enjoys reading, going out with friends, or watching movies. She has hypertension controlled with labetalol. Her blood pressure is 128/85 mm Hg. Physical examination shows a coarse tremor of the head and upper extremities. Deep tendon reflexes are increased. On mental status examination, she appears dysphoric and tearful. She is ruminative and repeatedly states "I am never going to get better." She wrings her hands and shifts her position in her chair throughout the examination. On cognitive testing, she answers almost every question with "I don't know." Results of laboratory studies are within the reference ranges. Which of the following is the most likely diagnosis?
- ☐ A) Bipolar disorder, depressed
 - ☐ B) Dementia
 - ☐ C) Generalized anxiety disorder
 - ☐ D) Late-onset schizophrenia
 - ☐ E) Major depressive disorder



8. A 32-year-old man comes to the office to request medication for a 6-year history of anxiety, worry, and insomnia. Three to four nights weekly, he lies in bed, unable to fall asleep and worrying about his life. He has no history of serious illness and takes no medications. He drank six beers daily for 10 years until he was convicted of driving while intoxicated 5 years ago. He now drinks only occasionally and in moderation. Physical examination shows no abnormalities. On mental status examination, he has a mildly anxious mood and congruent affect. There is no evidence of suicidal ideation or hallucinations. Results of complete blood count and serum thyroid-stimulating hormone concentration are within reference ranges. Which of the following is the most appropriate next step in management?
- ☐ A) Cognitive behavioral therapy
 - ☐ B) Polysomnography
 - ☐ C) Propranolol therapy
 - ☐ D) Quetiapine therapy
 - ☐ E) Recommendation for a 12-step program



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Pause

9. A 45-year-old man comes to the clinic because of a 4-month history of feeling tired in the morning and having difficulty concentrating. He reports feeling sleepy throughout the day and often dozes during important meetings. He has no trouble falling asleep at night and sometimes hears music while falling asleep. The patient's wife states that he typically sleeps soundly. He returned 2 weeks ago from a 15-day-long business trip to China. He is 170 cm (5 ft 7 in) tall and weighs 73 kg (160 lb); BMI is 25 kg/m². Blood pressure is 140/90 mm Hg; remaining vital signs are within normal limits. Physical examination shows no abnormalities. Results of sleep studies obtained 1 week later show the patient enters REM sleep within 5 minutes of two naps. Which of the following is the most likely diagnosis?
- ☐ A) Circadian rhythm sleep-wake disorder
 - ☐ B) Insomnia disorder
 - ☐ C) Narcolepsy
 - ☐ D) REM sleep behavior disorder





10. A 17-year-old girl is brought to the physician by her mother because of a 2-month history of difficulty sleeping and generalized fatigue. During this time, she also has had poor appetite, constipation, and dry skin. She is in high school and previously maintained an A average, but her grades recently declined to C's and D's. She states that she falls asleep in class at least once daily and has begun to skip morning classes so she can stay home and sleep. Her parents report that she no longer seems to have an interest in anything. She does not smoke cigarettes. She is 165 cm (5 ft 5 in) and weighs 57 kg (126 lb): BMI is 21 kg/m². Her temperature is 37°C (99°F), pulse is 62/min, respirations are 10/min, and blood pressure is 102/83 mm Hg. Physical examination shows pinpoint pupils and dry mucous membranes. The remainder of the examination shows no abnormalities. During the examination, she appears indifferent. Urine toxicology screening is most likely to show which of the following findings in this patient?

	Amphetamines	Opiates	PCP (phencyclidine)
☐ A)	Negative	negative	positive
☐ B)	Negative	positive	negative
☐ C)	Negative	positive	positive
☐ D)	Positive	negative	negative
☐ E)	Positive	negative	positive
☐ F)	Positive	positive	negative

11. An 18-year-old woman is brought to the emergency department by paramedics 30 minutes after her father found her lethargic and confused, with an empty bottle of "sleep aids" next to her. The patient has no history of serious illness, and her father was not aware that she was taking sleep medication. On arrival, the patient is agitated and uncooperative. She is not oriented to person, place, or time. Her temperature is 37.8°C (100°F), pulse is 128/min, respirations are 28/min, and blood pressure is 138/88 mm Hg. Physical examination shows 5-mm symmetric pupils and dry oral mucosa. The abdomen is soft and nontender; bowel sounds are decreased. Mental status examination shows decreased attention, intermittent somnolence, and agitation. She appears to be responding to internal stimuli and repeatedly screams at someone who is not in the room. Which of the following is the most likely cause of this patient's condition?
- ☐ A) Catecholaminergic blockade
 - ☐ B) Catecholaminergic excess
 - ☐ C) Cholinergic blockade
 - ☐ D) Cholinergic excess
 - ☐ E) Serotonergic blockade
 - ☐ F) Serotonergic excess



12. Over the past 2 months, a 28-year-old mother has had recurrent thoughts about stabbing her 2-year-old daughter. She has a visual image of attacking her daughter which provokes extreme anxiety. She is horrified by these violent thoughts but cannot prevent them from happening. She says she would never harm her daughter. She believes that she is losing her mind. She has no history of violence and says that she has not been depressed. Which of the following is the most appropriate pharmacotherapy for this patient?
- ☐ A) Clozapine
 - ☐ B) Fluoxetine
 - ☐ C) Imipramine
 - ☐ D) Lithium carbonate
 - ☐ E) Valproic acid



13. A 26-year-old woman, who was diagnosed with asthma 16 years ago, comes to the office for a routine examination. The patient reports that her boyfriend of 2 years broke up with her 1 month ago. She was surprised and disappointed, but she has kept busy since that time by taking a pottery class, volunteering with a suicide hotline, and planning a trip. Medical history also is remarkable for an episode of major depressive disorder 3 years ago that was treated with sertraline; she stopped taking this medication 2 years ago. Her only current medication is beclomethasone; she took an oral contraceptive for 5 years but discontinued it 4 weeks ago because she is not interested in dating at this time. She appears cheerful. Vital signs are within normal limits. Physical examination shows no abnormalities. This patient's behavior is most likely indicative of which of the following?
- ☐ A) Denial
 - ☐ B) Hypomania
 - ☐ C) Projection
 - ☐ D) Regression
 - ☐ E) Sublimation



14. A 63-year-old man is admitted to the hospital because of an ulcer on his left foot that has been enlarging during the past 3 weeks. The patient has a 15-year history of type 2 diabetes mellitus and is not adherent to his prescribed insulin regimen. He has no history of psychiatric illness. The patient is unable to provide information about his glucose concentrations at home. Evaluation confirms a foot infection, and amputation is recommended. The patient refuses the procedure, and a capacity evaluation is conducted. The patient reiterates that he does not want the amputation. The patient states that he has been told that the diabetes has led to an infection in the foot and that the blood flow is not strong enough to help his body fight the infection. He reports that he is receiving antibiotics and would like to continue them, even though the surgeons have told him that the chance for recovery is near zero. When asked what he thinks the most likely outcome of his choice will be, he responds, "I think the foot will get better. I have had diabetes a long time, and this has never happened before. I think the doctors are overreacting." Cognitive examination shows that he has intact attention, memory, and orientation to person, place, and time. Which of the following elements of capacity does this patient fail to demonstrate?
- ☐ A) Appreciation of the consequences of his choice
 - ☐ B) Communication of a choice
 - ☐ C) Reasoning about options
 - ☐ D) Understanding the relevant information



15. A 57-year-old man comes to the emergency department because of a 2-day history of moderate diarrhea and increasing clumsiness. He says he can no longer hold a coffee cup without spilling its contents. Four days ago, he saw a new physician, who started him on hydrochlorothiazide therapy for moderate hypertension, omeprazole for gastroesophageal reflux disease, lovastatin for hyperlipidemia, acetaminophen for back pain, and desloratadine for seasonal allergies. He has a 15-year history of bipolar disorder well controlled with lithium carbonate. He does not smoke, drink alcohol, or use illicit drugs. He does not appear dehydrated. His temperature is 37°C (98.6°F), pulse is 80/min, respirations are 12/min, and blood pressure is 140/105 mm Hg. Examination shows a fine tremor of both hands. His gait is mildly ataxic. On mental status examination, he is anxious but cooperative. He is oriented to person, place, and time. His serum electrolyte concentrations are within the reference range. His serum lithium concentration is 1.8 mEq/L (therapeutic range=0.6–1.2). Addition of which of the following medications most likely contributed to this patient's diarrhea and clumsiness?
- ☐ A) Acetaminophen
 - ☐ B) Desloratadine
 - ☐ C) Hydrochlorothiazide
 - ☐ D) Lovastatin
 - ☐ E) Omeprazole

16. An 87-year-old man comes to the office because of intermittent memory loss during the past 6 months. On several occasions, he has forgotten where he parked his car at the mall. Once, he eventually found the car while walking around; another time, a parking lot attendant drove him around until he found it. He has noticed no other changes in his functioning, but he is concerned because he prides himself on his ability to recall even insignificant details. He has migraines for which he takes intranasal sumatriptan as needed and ibuprofen. He appears to be in good health. Physical examination shows no abnormalities. On mental status examination, he is pleasant and cooperative. His mood is anxious, and he has full range of affect. He is fully oriented. He recalls one of three words after 1 minute. After the examination is complete, he suddenly recalls the other two words. He says that this happens frequently, and it is his inability to retrieve memories on demand that is frustrating. Which of the following is the most likely explanation for this patient's memory loss?
- ☐ A) Dementia, Alzheimer type
 - ☐ B) Dissociative amnesia
 - ☐ C) Illness anxiety disorder
 - ☐ D) Pseudodementia
 - ☐ E) Normal aging



17. A 20-year-old woman is brought to the emergency department after her roommate found her in her dormitory room next to an empty unlabeled pill bottle. Administration of 100% oxygen by face mask and intravenous naloxone was begun en route to the hospital. On arrival, her temperature is 35.6°C (96°F), pulse is 100/min, respirations are 6/min and shallow, and blood pressure is 94/60 mm Hg. She is moderately responsive to painful stimuli. Her serum glucose concentration is 68 mg/dL. Arterial blood gas analysis shows:

pH	7.19
Pco ₂	65 mm Hg
Po ₂	108 mm Hg
HCO ₃ ⁻	20 mEq/L

Results of toxicology screening are pending. Which of the following is the most appropriate next step in management?

- ☐ A) Delay treatment until results of toxicology screening are available
- ☐ B) Intravenous administration of 50% dextrose in water
- ☐ C) Intravenous administration of sodium bicarbonate
- ☐ D) Administration of activated charcoal by nasogastric tube
- ☐ E) Intubation and mechanical ventilation



18. A 27-year-old man comes to the physician 2 days after being treated in the emergency department for his third episode of chest pain, shortness of breath, nausea, and fear of dying over the past 2 weeks. Examination shows normal findings. Alprazolam therapy is initiated, and a follow-up examination is scheduled. This patient should be monitored for which of the following at his return visit?

- ☐ A) Agranulocytosis
- ☐ B) Akathisia
- ☐ C) Orthostatic hypotension
- ☐ D) Rebound anxiety
- ☐ E) Urinary retention



19. A 7-year-old boy is brought to the physician by his social worker 3 days after being placed in his fourth foster home in the past year. His social worker states that his behavior is making it difficult to find foster care placements for the boy, and he is at risk for placement in a group home. In each home, he becomes easily frustrated, aggressive, and is disrespectful. He has hit and injured younger children in some of the homes. He does not sleep well, cries out at night, and has nightmares. He refuses to go to school and sometimes hides under his bed. The social worker does not know details about his early life except that both of his parents were active substance abusers, and he and his older sister were found in an abandoned apartment with no heat, water, or food when he was 4 years old. He is at the 40th percentile for height and the 30th percentile for weight. Physical examination shows several healing abrasions over both knees. On mental status examination, he is irritable, easily frustrated, and impulsive. Which of the following is the most likely diagnosis?

- A) Acute stress disorder
- B) Oppositional defiant disorder
- C) Post-traumatic stress disorder
- D) Separation anxiety disorder
- E) Social anxiety disorder (social phobia)

20. A 94-year-old woman is brought to the office by her daughter for a follow-up examination. Three days ago, the patient was diagnosed with a urinary tract infection and prescribed oral antibiotics, which she has refused to take. She also has refused to eat and has been drinking minimal fluids during this time. She has dementia, Alzheimer type and lives in a nursing care facility. For the past 3 months, she has been unable to ambulate, get dressed, or bathe without assistance. Her verbal communication is limited to one-word answers. The patient's daughter has paperwork indicating that she is her mother's healthcare proxy; she says that her mother does not want extraordinary life-sustaining measures, including parenteral nutrition and hydration. Which of the following is the most appropriate next step in management?
- ☐ A) Intravenous administration of fluids and antibiotics
 - ☐ B) MRI of the brain
 - ☐ C) Nasogastric administration of fluids and antibiotics
 - ☐ D) Percutaneous endoscopic gastrostomy
 - ☐ E) Periodically offering semi-solid food by mouth



21. A 32-year-old woman is brought to the emergency department by her husband after the sudden onset of blindness. She has been under the care of a psychiatrist for the past 18 months and has been treated with clonazepam and fluoxetine. She has had no changes in sleep or appetite. Yesterday, she witnessed an elderly woman being struck and killed by a bus. On physical examination, the pupils are equal and reactive to light. There is bilateral nystagmus when the physician moves a strip of paper with vertical stripes in front of her eyes. Funduscopic examination shows no abnormalities. She has an unsteady gait and takes small steps with her arms outstretched. Which of the following is the most likely diagnosis?
- ☐ A) Adjustment disorder
 - ☐ B) Acute stress disorder
 - ☐ C) Factitious disorder
 - ☐ D) Functional neurologic symptom disorder
 - ☐ E) Somatic symptom disorder



22. A 32-year-old man comes to the emergency department because of a 4-hour history of worsening abdominal pain. He also has felt nauseated and "bloated" for the past day, and during the past 2 hours, he has had three episodes of emesis. The patient has bipolar disorder and has been treated with valproic acid for the past 4 years. His only other medication is ibuprofen for pain as needed. He has smoked one pack of cigarettes daily for the past 15 years and drinks ten glasses of wine weekly. He does not use illicit drugs. His diet consists primarily of fast food. The patient appears very uncomfortable and is curled into a ball on the examination table. Temperature is 37.2°C (99.0°F), pulse is 82/min, respirations are 18/min, and blood pressure is 148/92 mm Hg. Physical examination shows diaphoresis. There is significant tenderness to palpation of the middle and upper left abdomen. Results of laboratory studies are pending. Which of the following is the most likely cause of this patient's symptoms?
- ☐ A) Alcohol consumption
 - ☐ B) Cigarette smoking
 - ☐ C) Fast food consumption
 - ☐ D) Ibuprofen
 - ☐ E) Valproic acid



23. A 47-year-old man is brought to the emergency department by police because of extreme confusion and unintelligible communication; he was found wandering in a street. On arrival, he requires restraints for combativeness. He repeatedly yells, "Get them off me!" He is uncooperative and unable to give any history. His temperature is 37.5°C (99.5°F), pulse is 112/min, respirations are 24/min, and blood pressure is 156/104 mm Hg. Examination shows a thin, disheveled man with normal pupils. There is palmar erythema and gynecomastia. Mental status examination shows he is not oriented to person, place, or time, and he refers to the physicians as sailors. Serum studies show a glucose concentration of 72 mg/dL, total protein concentration of 5.5 g/dL, and albumin concentration of 3.0 g/dL. Toxicology screening is negative. Which of the following is the most likely diagnosis?
- ☐ A) Brief psychotic disorder
 - ☐ B) Delirium, alcohol withdrawal
 - ☐ C) Hypercalcemia
 - ☐ D) Schizophrenia
 - ☐ E) Withdrawal, cannabis



24. A 42-year-old man with widely metastatic lung cancer is admitted to the hospice unit. He has documented metastatic disease to liver and bone, and it has been determined that further attempts to treat his cancer would be futile. The patient has a history of hepatitis C and intravenous drug use; he reports last using heroin 6 months ago. Treatment of severe bone pain with fentanyl patches has not been effective. The patient is alert, fully oriented, and appears uncomfortable. Which of the following is the most appropriate approach to pain management for this patient?
- ☐ A) Add buprenorphine for breakthrough pain
 - ☐ B) Add short-acting opioids for breakthrough pain to the current regimen
 - ☐ C) Continue to use only long-acting opioid medications for pain
 - ☐ D) Discontinue fentanyl and initiate buprenorphine
 - ☐ E) Discontinue fentanyl, initiate methadone, and avoid short-acting opioids



25. A 72-year-old nursing home resident is brought to the physician because of paranoid behavior and irritability since she moved in 1 year ago. She reports poor sleep and believes that her telephone is being tapped and has refused to interact with her family. She reports hearing voices and believes she is being plotted against. Medications include haloperidol. She has a history of alcohol use disorder and was hospitalized for psychiatric symptoms 30 years ago, but records are unavailable. She appears agitated and irritable. Physical examination shows no abnormalities. Mental status examination shows a neutral mood and auditory hallucinations. Her insight and judgment are poor. An MRI of the brain shows age-related changes but is otherwise normal. Which of the following is the most likely diagnosis?
- ☐ A) Alcohol-induced psychotic disorder
 - ☐ B) Bipolar disorder
 - ☐ C) Generalized anxiety disorder
 - ☐ D) Major depressive disorder
 - ☐ E) Psychotic disorder due to a general medical condition
 - ☐ F) Schizophrenia
 - ☐ G) Schizotypal personality disorder



26. A 25-year-old woman comes to the office for a routine examination. During the interview, she says she has had a lack of energy, occasional anxiety, and difficulty sleeping during the past 2 to 3 years. She says she no longer takes pleasure in activities that she used to enjoy. She smokes marijuana twice monthly but does not use any other illicit drugs. She is 173 cm (5 ft 8 in) tall and weighs 68 kg (150 lb); BMI is 23 kg/m². Her pulse is 68/min, and blood pressure is 110/70 mm Hg. Physical examination shows no abnormalities. Results of laboratory studies are shown:

Hematocrit	40%
Erythrocyte count	4.2 million/mm ³
Leukocyte count	9000/mm ³
Serum	
Na ⁺	140 mEq/L
Cl ⁻	100 mEq/L
K ⁺	4.0 mEq/L
Thyroid-stimulating hormone	3.5 mIU/L
Thyroxine (T ₄)	5 µg/dL

Urine toxicology screening is negative. Which of the following is the most likely diagnosis?

- ☐ A) Generalized anxiety disorder
- ☐ B) Hypothyroidism
- ☐ C) Major depressive disorder
- ☐ D) Persistent depressive disorder (dysthymia)
- ☐ E) Substance-induced depressive disorder



27. A 45-year-old man comes to the office for a routine follow-up examination. The patient reports that he feels "thirsty all the time," and has experienced frequent urination during the past 6 months. He does not find these symptoms overly bothersome and states that he drinks "a lot of water during the day." He has hypertension, type 2 diabetes mellitus, hyperlipidemia, hepatitis C, and bipolar disorder. He has not experienced any symptoms of mania or depression during the past 12 months. Medications are metoprolol, metformin, atorvastatin, and lithium carbonate. Vital signs are within normal limits, and physical examination shows no abnormalities. Mental status examination shows a euthymic mood. Results of laboratory studies are shown:

Hemoglobin A _{1c}	5.9%
Serum	
Urea nitrogen	15 mg/dL
Glucose, fasting	85 mg/dL
Creatinine	0.9 mg/dL
Na ⁺	140 mEq/L
AST	32 U/L
Lithium	0.9 mEq/L (therapeutic range=0.6–1.2)

Which of the following is the most appropriate next step in addressing this patient's symptoms?

- ☐ A) Discontinue lithium carbonate and initiate divalproex
- ☐ B) Encourage continued drinking of water
- ☐ C) Increase dose of metformin
- ☐ D) Initiate desmopressin
- ☐ E) Recommend a low-protein diet
- ☐ F) Recommend fluid restriction

28. A 25-year-old woman is brought to the emergency department by friends after having a brief generalized tonic-clonic seizure. She is alert and restless and complains of a generalized headache. Her pulse is 110/min. She has signs of mild rhinitis and a small laceration of the tongue. Which of the following is the most likely substance taken?
- ☐ A) Alcohol
 - ☐ B) Barbiturate
 - ☐ C) Cocaine
 - ☐ D) Heroin
 - ☐ E) LSD
 - ☐ F) Marijuana
 - ☐ G) PCP

29. A 28-year-old woman is admitted to the hospital through the emergency department (ED) because of a 3-day history of weakness, fatigue, and occasional palpitations. In the ED, serum potassium concentration was 2.1 mEq/L. The patient says she is dissatisfied with her body image, leading her to periodically purge and secretly take her mother's hydrochlorothiazide. Her mood has become increasingly depressed since a long-term relationship ended 2 months ago. She has withdrawn from her friends, set aside her painting hobby, and often retreats to the restroom at work to cry privately. She is anxious she will be fired from her job as a barista in a coffee shop because poor sleep has contributed to her making multiple errors in completing customer orders. The patient appears weary but makes eye contact. She is 167 cm (5 ft 6 in) tall and weighs 73 kg (161 lb); BMI is 26 kg/m². Temperature is 36.7°C (98.0°F), pulse is 86/min, respirations are 12/min, and blood pressure is 108/67 mm Hg. Physical examination discloses no abnormalities. On mental status examination, the patient speaks softly and has a dysphoric mood and distressed affect. Thought process is linear. There is no evidence of suicidal ideation. Which of the following medications is contraindicated for this patient?
- ☐ A) Bupropion
 - ☐ B) Clonazepam
 - ☐ C) Desvenlafaxine
 - ☐ D) Escitalopram
 - ☐ E) Selegiline

30. A 20-year-old man comes to the physician at the request of his parents. He states that his parents are concerned because he was recently expelled from college for repeatedly violating dormitory rules. They do not understand how this could happen to him, because he graduated in the top third of his high school class. He is currently not working and does not have a permanent residence. He drinks three beers nightly and smokes marijuana occasionally. Physical examination shows no abnormalities. On mental status examination, he is irritable. There is no evidence of depression, psychosis, or suicidal or homicidal ideation. During the interview, he says that he does not want any sort of treatment or to come back to the clinic and does not want his parents contacted. He says that he is an adult and has the right to make these choices, even if they get him into trouble. Which of the following is the most appropriate next step in management?
- ☐ A) Abide by the patient's wishes
 - ☐ B) Explain to him that if he does not cooperate, his parents can have him admitted to the psychiatric unit
 - ☐ C) Schedule a family meeting and tell the patient you expect to see him there
 - ☐ D) Schedule individual psychotherapy
 - ☐ E) Tell the patient you will arrange for residential substance abuse treatment



31. A 27-year-old man comes to the physician with his wife at the insistence of his employer because of a recent episode of disruptive behavior. He works as a software engineer. His employer reports that the patient began shouting at another employee in the cafeteria for no apparent reason. The patient states that "My boss thinks I'm stressed out and maybe I am, but I know what's really going on there." He says that the company is constantly spying on him. He states that company operatives are opening and reading all of his mail, that his phone is bugged at work and at home, and that hidden cameras are watching his every move. He adds that the company wants to steal his ideas about a new type of computer game. He has no history of serious illness and takes no medications. He drinks one to two beers daily. Physical examination shows no abnormalities. On mental status examination, he says that he has intermittent anxiety about the company spying on him. He says he has not had hallucinations, depressed mood, difficulty sleeping, or decreased appetite. His wife notes that he has increased vigilance and a tendency to examine his mail carefully, but she reports no other changes in his behavior. Which of the following is the most likely diagnosis?
- ☐ A) Bipolar disorder
 - ☐ B) Delusional disorder
 - ☐ C) Generalized anxiety disorder
 - ☐ D) Major depressive disorder
 - ☐ E) Schizophrenia

32. A 27-year-old primigravid woman at 10 weeks' gestation is brought to the emergency department by her husband because of bizarre behavior. He reports that during the past week, she has been sleeping only 4 to 5 hours nightly and has not slept at all for the past 2 days. She spends most of her time talking on the telephone and this morning began talking about having a special mission. She tried to leave their home naked but he stopped her. She has no history of psychiatric illness. She takes no medications and does not use illicit drugs. Prior to the pregnancy, she drank alcohol socially, but she has been abstinent for the past 2 months. Physical examination shows no abnormalities. On mental status examination, she is agitated and irritable. She is alert and oriented. Her speech is rapid and labile. She says she hears the voice of God speaking to her and describes receiving a special mission to end world poverty, which she must present to the Secretary General of the United Nations. Her concentration is impaired. Which of the following is the most appropriate next step in management?
- ☐ A) Carbamazepine therapy
 - ☐ B) Cognitive behavioral therapy
 - ☐ C) Electroconvulsive therapy
 - ☐ D) Supportive psychotherapy
 - ☐ E) Valproic acid therapy



33. A 24-year-old woman comes to the office because of a 6-month history of an inability to reach orgasm. She says that while her partner has been patient and does not pressure her, she is concerned that he will eventually break up with her. She has no significant medical history. She has taken drospirenone-ethinyl estradiol for the past 4 years. When asked about her sexual history, the patient becomes tearful as she describes a previous boyfriend who "coerced her into some experimentation that she did not enjoy." She drinks four to five 12-oz beers on weekends. She does not smoke cigarettes or use illicit drugs. Vital signs are within normal limits. Physical examination discloses no abnormalities. On mental status examination, she describes her mood as "frustrated," and her affect is tearful. She has not had any suicidal or homicidal ideation. There are no signs of psychosis. Which of the following is the most likely diagnosis?
- ☐ A) Adverse effect of drospirenone-ethinyl estradiol
 - ☐ B) Alcohol use disorder
 - ☐ C) Female orgasmic disorder
 - ☐ D) Major depressive disorder
 - ☐ E) Post-traumatic stress disorder



34. A 27-year-old woman comes to the physician because of generalized muscle and joint pain, nausea, and abdominal cramping for 36 hours. She has a history of chronic pelvic pain and has seen several different physicians about her condition. Her temperature is 37.2°C (99°F), respirations are 20/min, and blood pressure is 150/90 mm Hg. Examination shows excessive tearing, rhinorrhea, and piloerection. The most likely cause of these symptoms is withdrawal from which of the following?
- ☐ A) Alcohol
 - ☐ B) Amitriptyline
 - ☐ C) Butalbital
 - ☐ D) Diazepam
 - ☐ E) Hydromorphone



35. A 5-year-old girl is brought to the physician at the request of her teacher because of hyperactivity and learning difficulties since starting kindergarten. She was born at term by an uncomplicated spontaneous vaginal delivery but had a period of increased irritability and tremulousness soon after birth. She is at the 20th percentile for height and the 30th percentile for weight. Examination shows short palpebral fissures, an upturned nose, a hypoplastic philtrum, and a cardiac murmur. She has an IQ of 60. Which of the following is the most likely cause of her condition?
- ☐ A) Exposure to alcohol in utero
 - ☐ B) Exposure to cocaine in utero
 - ☐ C) Exposure to lead
 - ☐ D) Exposure to *Treponema pallidum* in utero
 - ☐ E) Fragile site on the X chromosome
 - ☐ F) Trisomy of a chromosome



36. A 72-year-old man is brought to the physician by his wife because of odd behavior and increasing difficulty completing simple tasks during the past 6 months. His wife says that he has begun speaking oddly at times and mixing his words. He says that he feels fine, has a good appetite, and sleeps well. Physical examination shows no abnormalities. On mental status examination, he is smiling, and he states that his mood is "okay." His affect is mildly anxious. He is oriented to time but not to person or place. He is able to recall zero of three objects after 5 minutes. He is able to name several presidents during the past 2 decades but not in the order of their presidency. Laboratory findings are within the reference range. Urine toxicology screening is negative. Which of the following is the most appropriate initial pharmacotherapy?
- ☐ A) Buspirone
 - ☐ B) Donepezil
 - ☐ C) Fluoxetine
 - ☐ D) Olanzapine
 - ☐ E) Vitamin B₁ (thiamine)



37. A 39-year-old woman is examined in the hospital 1 day after admission because she was found by her son in an unresponsive state with an empty bottle of topiramate nearby. She says she overdosed with suicidal intent because she has felt "lost" since completing radiation therapy for a left frontoparietal oligodendroglioma 3 months ago. She has felt anxiety about her future and has been ruminating about past mistakes in her life. She tosses and turns nightly and feels that each day is burdensome. During this time, she has had decreased appetite resulting in a 7-kg (15-lb) weight loss. Her father and brother have bipolar disorder. The patient is 165 cm (5 ft 5 in) tall and weighs 48 kg (105 lb); BMI is 18 kg/m². Vital signs are within normal limits. Physical examination discloses no abnormalities. On mental status examination, the patient's eyes are downcast, and she speaks with a soft voice. Her speech is scant and she exhibits psychomotor slowing. She has a depressed mood and restricted affect. Results of thyroid function tests are within the reference ranges. Following transfer to the inpatient psychiatry unit, which of the following is the most appropriate pharmacotherapy for this patient?
- ☐ A) Bupropion
 - ☐ B) Lithium carbonate
 - ☐ C) Lorazepam
 - ☐ D) Methylphenidate
 - ☐ E) Mirtazapine

38. An 8-year-old boy is brought to the physician because of increasing problems since entering second grade. Although he repeated first grade, he continues to have difficulty reading and writing. His teacher says that he is pleasant and well behaved but is immature and often teased by the other children. His sister has attention-deficit/hyperactivity disorder. He is at the 10th percentile for height and 75th percentile for weight. On examination, he is slightly fidgety but cooperative and appears well. His speech is rapid. He has a large head with a prominent forehead and jaw and large protruding ears. The remainder of the examination shows no abnormalities. Which of the following is the most likely explanation for this patient's learning problems?
- ☐ A) Chemical imbalance
 - ☐ B) Chromosomal abnormality
 - ☐ C) Environmental toxin
 - ☐ D) Enzyme deficiency
 - ☐ E) Intrauterine teratogen



39. A 77-year-old man is admitted to the hospital because of a 2-week history of insomnia, agitation, and auditory hallucinations. He says he often hears accusatory voices telling him that he is "no good." He has Parkinson disease that has been treated with carbidopa-levodopa therapy for the past 3 years. He has no history of psychiatric illness. Physical examination shows a rhythmic resting tremor that improves on intention and mild rigidity. The patient frequently changes position during the examination. On mental status examination, he appears agitated. He has an anxious mood and labile affect. He reports continued auditory hallucinations. Which of the following is the most appropriate next step in management?
- ☐ A) Add citalopram to the medication regimen
 - ☐ B) Add haloperidol to the medication regimen
 - ☐ C) Add lorazepam to the medication regimen
 - ☐ D) Add olanzapine to the medication regimen
 - ☐ E) Decrease the dose of carbidopa-levodopa
 - ☐ F) Increase the dose of carbidopa-levodopa



40. A 29-year-old man with asthma comes to the clinic for a routine follow-up visit. The patient says that his girlfriend encouraged him to discuss potential treatments for his 7-month history of sexual difficulties. He says that his girlfriend is his first long-term relationship, and since they began dating, he has had premature ejaculation during sexual activity, with ejaculation occurring within approximately 30 seconds of vaginal penetration. They have tried various self-help strategies without improvement. The patient says that he has begun to avoid sexual activity because of embarrassment. He is 178 cm (5 ft 10 in) tall and weighs 66 kg (146 lb); BMI is 21 kg/m². Temperature is 36.9°C (98.4°F), pulse is 56/min, respirations are 16/min, and blood pressure is 112/72 mm Hg. Physical examination shows no abnormalities. Results of laboratory studies are within the reference ranges. Which of the following is the most appropriate pharmacotherapy at this time?
- ☐ A) Buspirone
 - ☐ B) Paroxetine
 - ☐ C) Sildenafil
 - ☐ D) Testosterone
 - ☐ E) Trazodone

41. A 37-year-old woman is brought to the emergency department by her husband 1 hour after the onset of palpitations, hyperventilation, dizziness, and light-headedness. On arrival, she is alert, fully oriented, and mildly anxious. Her husband says she has had similar episodes weekly during the past year and was seen in the emergency department following an episode 2 months ago. Examination at that time showed no abnormalities. He says his wife is reluctant to leave the house without him because she is fearful of having an episode in public. She prefers to shop for groceries late at night when few people are around. Physical examination shows a grade 3/6, systolic ejection murmur. ECG shows sinus tachycardia. Which of the following is the most likely diagnosis?
- ☐ A) Meniere disease
 - ☐ B) Mitral valve prolapse
 - ☐ C) Panic disorder
 - ☐ D) Paroxysmal atrial tachycardia
 - ☐ E) Schizoid personality disorder
 - ☐ F) Social anxiety disorder (social phobia)



42. A 61-year-old woman with hypertension and hyperlipidemia is brought to the clinic by her daughter because of a 4-day history of behavioral changes. The patient's daughter states that the patient has recently been awake much of the night, paranoid that people are coming into the house and poisoning her food. The daughter also says that her mother appears to be talking to herself and thinks she may be hearing things, although the patient has not reported auditory hallucinations. The patient is calm and pleasant, and says that her daughter is overreacting with her concerns. Medications are hydrochlorothiazide and rosuvastatin. She drinks one to two glasses of wine daily and does not use any illicit drugs. She is oriented to person and place but not to time. She appears restless. Temperature is 36.7°C (98.1°F), pulse is 90/min, respirations are 15/min, and blood pressure is 138/81 mm Hg. Pulse oximetry on room air shows oxygen concentration of 97%. Physical examination shows no abnormalities. Mental status examination discloses impaired attention and concentration. Which of the following is the most likely diagnosis?
- ☐ A) Alcohol withdrawal
 - ☐ B) Brief psychotic disorder
 - ☐ C) Dementia with hallucinations
 - ☐ D) Transient ischemic attack
 - ☐ E) Unspecified delirium



43. A 39-year-old man comes to the office for follow-up of a 10-year history of major depressive disorder. The patient has had two previous major depressive episodes during the past 10 years that required treatment with sertraline but the patient states that he is "better." He has recently changed jobs from a night shift position to a day shift. He describes himself as a shy, overly cautious person, who is generally pessimistic about events or situations in his life. He says he is unable to assert himself, even when he knows he is right. He believes he is "empathic, hard-working, and loyal." He has few friends and has difficulty making new ones. He likes to be demonstrative with close friends and family and craves affection, but he has difficulty establishing close relationships because of his shyness. Family history is significant for major depressive disorder and alcohol use disorder in multiple relatives. The patient's sleep and appetite are normal. Vital signs are within normal limits and physical examination shows no abnormalities. Which of the following is the most appropriate management of this patient's condition?
- ☐ A) Cognitive behavioral therapy
 - ☐ B) Lorazepam
 - ☐ C) Motivational interviewing
 - ☐ D) Quetiapine
 - ☐ E) No management is indicated

44. An otherwise healthy 42-year-old man comes to the office at his wife's insistence because she can no longer tolerate his moodiness. He says he used to be happy, but he has not enjoyed "daily activities most of the time" during the past decade. He says that his wife is probably correct about his irritability. Six weeks ago, their 17-year-old son returned home after being confined to a youth detention center for 1 month. The patient and his wife have been having difficulty with their son since he returned. The patient says he frequently feels like he "won't be good enough" and has no hope that he will feel better. He is a computer programmer and consistently performs well on the job. He has had no changes in appetite or weight during the past 5 years. He had a depressed mood that was successfully treated from the ages of 19 to 20 years. He currently takes no medications. Physical examination shows no abnormalities. On mental status examination, he has a worried mood and blunted affect. His speech is soft. Results of laboratory studies, including serum thyroid-stimulating hormone concentration, are within the reference ranges. Which of the following is the most likely diagnosis?
- ☐ A) Adjustment disorder with depressed mood
 - ☐ B) Cyclothymic disorder
 - ☐ C) Dependent personality disorder
 - ☐ D) Major depressive disorder
 - ☐ E) Persistent depressive disorder (dysthymia)

45. A 28-year-old primigravid woman is brought to the emergency department at 9 weeks' gestation by her husband because of a 1-day history of confusion and worsening nausea and vomiting. The patient's severe nausea and vomiting began 4 weeks ago. The patient's husband reports that during the past 2 days, she has been increasingly confused and forgetful. He has heard her singing and talking to herself, which is out of character. The patient is 168 cm (5 ft 6 in) tall and weighs 70 kg (154 lb); BMI is 25 kg/m². Temperature is 37.0°C (98.6°F), pulse is 127/min, respirations are 18/min, and blood pressure is 94/53 mm Hg. Physical examination shows dry mucous membranes. Neurologic examination shows nystagmus; she is somnolent on examination. Gait is mildly ataxic. On mental status examination, the patient recalls one out of three objects after 5 minutes. The most likely cause of this patient's symptoms is a deficiency of which of the following?
- ☐ A) Folic acid
 - ☐ B) Vitamin B₁ (thiamine)
 - ☐ C) Vitamin B₆ (pyridoxine)
 - ☐ D) Vitamin B₁₂ (cobalamin)
 - ☐ E) Vitamin D



46. An 8-year-old boy is brought to the clinic by his parents because of a 2-year history of behavioral problems and declining academic performance. His parents say he has always been very active and has had difficulty concentrating and completing tasks at home and school. He is currently repeating the first grade and receives grades of F's in all subjects. His father also had difficulty focusing in school as a child. The patient was born at term following an uncomplicated pregnancy and spontaneous vaginal delivery. He has no history of serious illness and receives no medications. He drinks two glasses of iced tea daily. He is at the 50th percentile for height and 25th percentile for weight. His pulse is 80/min, respirations are 20/min, and blood pressure is 90/60 mm Hg. In the examination room, he jumps on and off the examination table and frequently interrupts his mother. Deep tendon reflexes are 2+ and symmetric bilaterally. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in management?
- ☐ A) Advise the parents to remove caffeine from the patient's diet
 - ☐ B) Ask the patient's parents and teachers to complete a standardized questionnaire about his behavior
 - ☐ C) Begin stimulant therapy
 - ☐ D) Order thyroid function tests
 - ☐ E) Reassure the patient's parents that his behavior is normal for his age



Previous



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Lab Values



Calculator



Review



Help



Pause

47. A 16-year-old boy is brought to the emergency department by police for suspected drug intoxication after his mother found him breaking the mirrors and windows in their home. When the police arrived on the scene, the patient was calm and appeared confused. His mother reports that he has no history of violent behavior. His grades began declining 4 months ago, and he dropped out of school 2 months ago. He has no history of serious medical illness. He appears disheveled. His pulse is 78/min, respirations are 22/min, and blood pressure is 110/76 mm Hg. Examination shows mild erythema surrounding the mouth. There is a fruity odor to the breath. His movements are slow, and his gait is unsteady. On mental status examination, he is oriented to person, place, and time, but he does not recall the violent behavior. His speech is slow. Intoxication with which of the following substances is the most likely cause of this patient's symptoms?
- ☐ A) Alcohol
 - ☐ B) Cocaine
 - ☐ C) Heroin
 - ☐ D) Marijuana
 - ☐ E) Methamphetamine
 - ☐ F) Nitrous oxide
 - ☐ G) PCP (phencyclidine)
 - ☐ H) Toluene

48. A 32-year-old man is brought to the physician by his wife because of a gradual change in his personality during the past 2 years. He has become more self-centered and inconsiderate of his family, friends, and coworkers. He also sometimes makes rude and inappropriate comments. The patient says that he has no family history of a neurologic or psychiatric disorder. However, his wife says that he and his family do not discuss their medical history; she is fairly certain that the patient's mother committed suicide at approximately 40 years of age. On mental status examination, his mood is euphoric, and he tells the physician that she is a good-looking woman. On physical examination, saccades are slowed. There are occasional jerky movements of the face and trunk muscles and of the upper extremities. These movements are more prevalent in the distal upper extremities than in the proximal upper extremities. Which of the following is most likely to establish the diagnosis?
- ☐ A) No further testing is required
 - ☐ B) Complete psychiatric assessment
 - ☐ C) Neuropsychological testing
 - ☐ D) Genetic testing
 - ☐ E) MRI of the brain
 - ☐ F) Electroencephalography

49. A 25-year-old woman is brought to the emergency department by her boyfriend 90 minutes after he witnessed her ingesting 25 100-mg amitriptyline tablets. She has not consumed any alcohol or illicit drugs. He reports that they were arguing about her continued interest in a previous boyfriend. She has a 10-year history of self-destructive behavior, including cutting herself. She is somnolent but easily arousable. Her temperature is 38°C (100.4°F), pulse is 108/min, respirations are 10/min, and blood pressure is 90/60 mm Hg. The pupils are 4 mm and reactive. On mental status examination, she is oriented to person, place, and time. She is intermittently loud and argumentative with her boyfriend when she is fully awakened. Which of the following abnormalities is most likely to be noted on this patient's ECG?
- ☐ A) QRS-complex prolongation
 - ☐ B) QRS-complex shortening
 - ☐ C) ST-interval prolongation
 - ☐ D) ST-interval shortening
 - ☐ E) U-wave prolongation
 - ☐ F) U-wave shortening



Previous



Next

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Lab Values



Calculator



Review



Help



Pause

50. A 9-year-old girl is brought to the office because of changes in behavior during the past 6 months. Her mother reports that her daughter has always been clean and organized but has become increasingly focused on cleanliness. She now takes very long showers, constantly cleans her room, bleaches her belongings, and frequently washes her hands. The patient reports that she is very aware of germs in the environment and cannot stop worrying about them. She performs well in school and receives grades of A's. She says that when she feels anxious at home or school, she has difficulty resisting the urge to wash her hands or clean her desk. Examination shows erythematous, excoriated, irritated skin over the dorsum of the hands. There is no bleeding or purulent discharge. No ulcers are noted. In addition to treating this patient's skin condition, which of the following is the most appropriate management for her underlying disorder?
- ☐ A) Buspirone therapy
 - ☐ B) Cognitive behavioral therapy
 - ☐ C) Family therapy
 - ☐ D) Risperidone therapy
 - ☐ E) Social skills training
 - ☐ F) Venlafaxine therapy

